The Spaghetti Engine
VISUAL SYSTEM MODELLING WORKSPACE

The Spaghetti Engine

Development Log

Fuzzy Cognitive Map modelling tool

Field	Value
Version	0.4.5 (prototype)
Log entry date	19 October 2026
Status	Released: deployed by the push of this release to the GitHub main branch (the commit titled Release 0.4.5), together with the 0.4.5 documents and this log. Netlify deploys the site automatically.
Live URL	spaghetti-engine.netlify.app (Netlify, auto-deployed from the GitHub main branch)
Source repository	github.com/aPaarl/Spaghetti-engine, main branch (Release 0.4.5)
Previous version	0.4.4, 14 October 2026, commit 7ba922c, documents in commit 25cb584 (see Spaghetti_Engine_Dev_Log_v0.4.4.docx)
Prepared by	A. Paarlberg

1. Purpose of This Log

This Development Log records the changes from version 0.4.4 (14 October 2026) to version 0.4.5 of The Spaghetti Engine. Version 0.4.5 reworks the Network tab so that a map can serve as a figure of the system: a Figure style with print-weight lines, lines routed around cards and labels kept clear of each other, a cropped high-resolution export, and four smart Auto Arrange layouts built on a new layout engine.

The log documents what changed and why, how existing results are affected, how the changes were checked, and the deployment status.

2. Summary of Version 0.4.5

Area	Change
Figure style	A Figure / Editing switch on the Network tab. Figure (the default): name-only cards, thin lines dashed when decreasing, no anchor dots or grid at rest. Editing: the 0.4.4 look.
Routing and labels	In Figure style, lines are straight and bow only around cards in their way or apart from a reciprocal partner; weight labels are placed along their line clear of cards and other labels, and scale with the map.
Export PNG	In Figure style: the whole map cropped to its content, at three times screen resolution, without controls, minimap, or anchor dots.
Smart Auto Arrange	Readable, Systems thinking, Cluster view, and Centrality view, from a new layout engine (Louvain groups, group and coarse force layouts, local search on a weighted readability objective), run in a Web Worker and animated into place as one undo step.
Classic layouts	Circular, force-directed, and grouped-by-module now push apart any cards they leave overlapping; the layered layouts size each card from its name in Figure style.
Record of 0.4.4	Version 0.4.4 was pushed as commit 7ba922c on 14 October 2026, with its documents in commit 25cb584.

3. The Changes in Detail

3.1 Figure style

- **Cards.** In Figure style a card shows only the concept's name (activation in its tooltip), with a 1.5-pixel module-coloured border and no shadow. The twelve connection dots are invisible until the card is hovered or a relationship is being drawn, and the background grid is hidden.
- **Lines.** Stroke width 1.1 + 2.2|w| (bundled links 1.4 + 0.45 per relationship, up to eight), against 2.25 + 5.5|w| in Editing style; arrowheads scale with it. Decreasing relationships are dashed, in proportion to the stroke width.

- **Routing.** A line that would pass through another card is bowed by the smallest offset (0, then 40, 80, 130, or 190 pixels, either side) that clears every card; reciprocal pairs and hand-made bends keep their own offsets on top. The bend is recomputed when cards stop moving, not during a drag.
- **Labels.** A weight label is a small monospace number on a white halo, placed at the fraction of its line (0.5, 0.42, 0.58, out to 0.27 or 0.73) that overlaps the fewest cards and labels already placed, strongest relationships first. In Figure style labels are sized in map space, so they scale with the cards when zooming.
- **Remembered.** The chosen style is stored in the browser, and the toolbar's Curved button reads Direct in Figure style.

3.2 Smart Auto Arrange

- **Engine.** src/layout/fcmLayout.js (new, framework-free) computes the node metrics, the Louvain groups with the Leiden connectivity guarantee, the group and coarse layouts, the rotation of groups, and the local search; Section 4.7 of the Methodological Annex gives the method and the objective.
- **Modes.** Readable (the default), Systems thinking (drivers left, outcomes right, short feedback loops as rings), Cluster view (the model's modules when it has two or more, otherwise detected groups, widely separated), and Centrality view (central concepts in the middle).
- **Running.** The engine runs in a persistent Web Worker (src/layout/fcmLayout.worker.js), falling back to the main thread if a worker cannot start. The result is centred on the current arrangement, the cards glide to it in 650 milliseconds (skipped when the system asks for reduced motion), and the model changes once at the end, as one undo step.
- **Report.** A line under the toolbar gives the crossings, overlapping cards, groups, feedback structures, and time; its tooltip gives the score and its terms.
- **Interim layout removed.** A Figure layout added during development, before this release, is replaced by Cluster view.

3.3 Export

- **Figure export.** In Figure style, Export PNG renders React Flow's viewport layer at scale 1, shifted so the bounds of all cards plus a 60-pixel margin fill the image, at a pixel ratio of 3, with the connection dots filtered out. The file is named spaghetti-engine-network-figure.png (with the scope added when a partial scope is active). Editing style keeps the 0.4.4 export.

4. Implementation Choices

Topic	Choice in the tool	Reason
Groups	Louvain plus the Leiden connectivity guarantee, not the full Leiden algorithm.	The refinement phase of Leiden mainly prevents badly connected communities; splitting each Louvain group into its connected parts gives that guarantee at a fraction of the code, and the groups only steer the drawing.

Topic	Choice in the tool	Reason
Search budget	A fixed amount of work, with a time limit only as a safety net.	A time budget would make the result depend on how busy the computer is; a work budget gives the same layout for the same model.
Layered seed for Systems thinking	Not used.	An ELK layered layout was tried as a starting point; on maps with large feedback components it neither reduced crossings nor saved time, and added about 0.5 seconds.
Language	JavaScript with JSDoc types.	The project has no TypeScript build; the types are checked by editors as is, and moving to TypeScript needs only a rename and a configuration.
Where labels sit	Chosen per line after placement, in Figure style only.	Label positions depend on the bends and on the measured card sizes, which the layout engine does not know; Editing style keeps the 0.4.4 midpoint labels.

5. Effect on Existing Results

- No analysis result changes: the Figure style and the layouts change only how the map is drawn.
- A saved model keeps its positions; they change only when Auto Arrange is used. The map opens in Figure style the first time 0.4.5 is used in a browser; Editing restores the 0.4.4 look.
- A network map exported in Figure style differs from one of 0.4.4: it shows the whole map at its own scale, not the current view.
- Version raised from 0.4.4 to 0.4.5 in package.json and package-lock.json.

6. Defects Found and Fixed

No defects of earlier releases were found or fixed in this release.

7. Verification

- **Tests:** tests/test_fcm_layout.mjs (new, 57 checks) tests Louvain on two bridged cliques, betweenness and eigenvector centrality on a chain, and the detection of a 3-cycle, and runs every mode on synthetic maps of 20, 50, 100, and 200 concepts with planted clusters, drivers, and loops. It checks that no cards overlap, that Readable has fewer crossings than a plain force-directed layout at every size (for example 37 against 46 at 50 concepts and 122 against 250 at 100), that the other modes stay within a stated margin of it, that drivers sit left of outcomes in Systems thinking, that the same input gives the same layout, and the time limits (scaled to the measured speed of the machine). tests/test_marginal_benefit_tp.cjs (37 checks), tests/test_transition_direction.cjs (37), tests/test_pathway_blank_inputs.cjs (5), and tests/test_pathway_expert_robustness.cjs (28) still pass.

- **In the app:** on the 40-concept water-energy-food example, Readable reduced the crossings (lines through cards included) from 514 in the example's own arrangement to about 100, in 0.4 seconds; every mode ran from the menu, animated into place, and reported its result, without console errors. The Figure style export of that model was checked at 5343 by 4719 pixels.
- **Timing:** on the build laptop, running about 1.4 to 2 times slower than usual during the tests, a smart layout took about 0.2 to 0.4 seconds for 20 concepts, 0.4 to 0.6 seconds for 50, about 1 second for 100, and 2 to 3 seconds for 200.
- **Build:** the production build completes without errors, with the worker as a separate 30 kB file.

8. Deployment Status

- Deployed with the push of the release commit to main on 19 October 2026, together with the 0.4.5 documents and this log. Netlify rebuilds the site automatically.
- **Files changed:** src/SpaghettiEngine.jsx, src/layout/fcmLayout.js (new), src/layout/fcmLayout.worker.js (new), tests/test_fcm_layout.mjs (new), docs/auto-arrange-design.md (new, the design note), package.json and package-lock.json; with the documents, src/docs-manifest.json, public/sitemap.xml, and the three PDFs in public/docs, where the 0.4.5 versions replace those of 0.4.4.
- **Build note:** npm run build fails in the local OneDrive folder because npm's command shim cannot handle the ampersand in the folder path; running node node_modules/vite/bin/vite.js build directly works.
- **PDF note:** as for 0.4.3 and 0.4.4, the PDFs were printed with Chrome (EDGE_PATH set to chrome.exe).

9. Prototype Scope and Known Limitations

- All limitations listed in the 0.4.0 to 0.4.4 logs still apply.
- Systems thinking keeps drivers left and outcomes right, so feedback running back across the map costs crossings: on the synthetic 50-concept map it had about twice as many as a plain force-directed layout.
- On maps of around 200 concepts with many relationships between groups, a smart layout still leaves many crossings (952 against 1215 for a plain force-directed layout on the synthetic test map); such maps are better read with modules folded.
- The layout time is not guaranteed: the search does a fixed amount of work, so a slow computer takes longer, up to the safety limit.
- The User Manual and the Methodological Annex still do not describe the expert-defined sequences and the three robustness families of 0.4.4; Sections 3.2 and 3.3 of the 0.4.4 log describe them.

10. Related Documentation

Document	Covers
The Spaghetti Engine - User Manual (version 0.4.5)	How to use every feature. Section 1.2 lists the changes of 0.4.5; Section 4.1 describes the Figure and Editing styles and Auto Arrange.

Document	Covers
The Spaghetti Engine - Methodological Annex (version 0.4.5)	The methods. Section 4.7 describes the automatic layout and the Figure style routing.
docs/auto-arrange-design.md (repository)	The design note of the layout engine: architecture, algorithm, results, and code changes.
Spaghetti_Engine_Dev_Log_v0.4.4	The 0.4.4 release: blank inputs, expert-defined sequences, and the robustness families.
This Development Log	What changed in 0.4.5 and the technical work behind it.

11. Next Steps

- Describe the expert-defined sequences and the three robustness families of 0.4.4 in the User Manual and the Methodological Annex.
- Consider highlighting the feedback loops found by the layout on the map itself, and a layered layout of the condensed graph for Systems thinking on maps with few feedback loops.
- Rebuild the full engine regression set inside tests/.
- Continue with the development roadmap: version 0.5.0 (guided first use) is next.